

M. PHARM. (PHARMACOLOGY)
(SEM II) THEORY EXAMINATION 2022-23
PRINCIPLES OF DRUG DISCOVERY

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**10 x 2 = 20**

- (a) What are microarrays?
- (b) Give examples of zinc finger proteins and its role in drug discovery.
- (c) Write the role of combinatorial chemistry in lead identification.
- (d) What are pharmacophores? Give examples.
- (e) Discuss de novo drug design.
- (f) Outline the role of NMR in protein structure prediction.
- (g) Explain the various structural levels of proteins.
- (h) Describe Lipinski rule of five.
- (i) What is SiRNA?
- (j) Discuss free Wilson analysis.

SECTION B

2. Attempt any two parts of the following:**2 x 10 = 20**

- (a) Describe the various stages of modern drug discovery process. How much economy is involved in bringing a new drug molecule in market.
- (b) Define QSAR. Explain various physicochemical properties in QSAR. Add a note on Hansch Analysis.
- (c) Discuss threading and homology modelling. Explain the prediction of protein structure by homology modelling.

SECTION C

3. Attempt any five parts of the following:**5 x 7 = 35**

- (a) Discuss the role of genomics in target discovery and utilization.
- (b) Write short note on Antisense oligonucleotides in target discovery and validation.
- (c) Explain high throughput screening technique for lead identification.
- (d) Illustrate the concept of Pharmacophore mapping in rational drug design.
- (e) What is flexible docking? Explain docking based screening approach in detail.
- (f) Discuss the 3D-QSAR technique CoMFA in detail.
- (g) Explain the role of prodrugs design with its applications.